

Element C

Presentation and Justification of Solution Design Requirements

Problem Statement:

In this project our goal is to “design a durable, lightweight, replaceable, secure mic flag that communicates and promotes the TimberTV and Ambush News’ logo and allows broadcasters to have multiple flag options to choose from when recording with quantifiable constraints such as weighing under 100 grams and an inner circle diameter of less than 1 ¼ inches.”

Our design requirements are as follows:

Design Requirement #1 (HIGH Priority)

A high priority requirement for our mic flag would be the logos on the side of the microphone not to fade, and last a long time. Mr O’Farrell felt like this was the most important design requirement as the main purpose of Mic Flags is to advertise. If the images fade, they are not as good at advertising as they would be when it is new. To test this design requirement, we will make 2 images, one will sit in direct sunlight for 24 hours while the other will be kept as a control. After one day we will compare the two, if fading is present, we will go to a new design.

Design Requirement #2 (HIGH Priority)

Another high priority is for the advertisements to have a large enough panel to advertise on. This directly ties to prior research because it is the purpose of the mic flag to show clearly present advertisements. To pass this design requirement, the advertisements need to be on an area of at least 75 by 50 millimeters. This test is objective and measurable as we will be able to measure without an opinion. Mr O’Farrell liked this idea as a panel that is smaller than that would have a tougher time advertising and would not look as good on the mic.

Design Requirement #3 (HIGH Priority)

The mic flag designs would have to be fairly durable if they came into contact with harsh conditions from the user. The mic flag has to be fairly durable because if the mic flag were to be dropped then it shouldn’t fall apart like legos because then it would destroy the flag

itself and maybe even hurt the mic a little bit. To test this we will drop our mic flag from a height of 1 foot. If the flag breaks, it fails the test. Mr O'Farrell said that this is important as some students can be clumsy with the mic and on occasions the flag drops.

Design Requirement #4 (HIGH Priority)

The mic flag will need to be replaceable when damaged. if the mic flag gets damaged in any way, then we would need to be prepared for this situation. We could have an extra set of flags or the flag base if it breaks in any way. To test this, we must be able to produce a new mic flag within 1 day. This includes everything from the printing to the addition of logos. This is important to Mr O'Farrell as he does not want to run out of mic flags if he has issues with the current ones and we cannot reproduce them in a timely manner.

Design Requirement #5 (HIGH Priority)

The mic flag will have to be able to go on multiple different kinds of microphones and fit on them. This design plays an important role because if the mic flag can only fit a certain amount of mic flags then there is only 1 mic that the flag could work on which could cause problems if that mic broke in some way. To test this, we will put the mic flag on 5 mics of different sizes. It will spend 5 minutes on each mic. After that, the flag will be shaken on each mic to make sure it stays attached and does not loosen so that it falls off of smaller mics. Mr O'Farrell would like to have mic flags that fit multiple mics so that he can have fewer mic flags, but it is not his biggest concern with the flags.

Design Requirement #6 (LOW Priority)

A low priority for the mic flag is for it to be not sharp and we will test that by pushing it against cardboard and seeing how much it rips through it. This creates safety for the user and others in case the flag hits someone or damages something and if it doesn't pierce cardboard at all, then it's safe. To test this we will drag the corner of the flag against a piece of cardboard, if it pierces the cardboard, it will be a failure for being too sharp. Mr O'Farrell isn't too concerned about the sharpness as the chance someone gets hit by the flag is low, but he agrees that the corners should not be overly sharp so that it pierce cardboard.

Design Requirement #7 (LOW Priority)

Another low priority for the Mic flag is that it can't weigh over 100 grams otherwise the mic flag could be too heavy then it could fall off the microphone. If we make the flag heavy then the flag will start to fall and slip off. It will also make the mic top heavy which will make it



tougher for the presenter with the mic to hold it. When we meet the requirements, the flag will be successful and won't slip off the mic and will not make it top heavy. To test this we will weigh our mic on a scale to see if it passes. We originally had this requirement higher up the list, but after talking to Mr O'Farrell, he felt that the weight of the flag would not be an issue as long as the flag is not overly heavy.